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Placement Pal: Placement Readiness & Assessment Platform .

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Project Guide
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Outline

- Introduction
- Objectives
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement
- System Design
- Technologies and methodologies
- Implementation
- Conclusion
- References

Sustainable Development Goals (SDG) mapped

SDG 9: Industry, Innovation, and Infrastructure

1.**Data-driveninsights:**
Directly highlights AI (LLMs + RAG) usage
Strongly aligns with innovation and advanced technology infrastructure
Clearly reflects the core technical strength of your project

2.**Infrastructure Skill Development:**
Emphasizes your platform as a digital infrastructure
Includes key features like AI generation, proctoring, and learning roadmaps

Introduction

- What is Placement Pal?
- A comprehensive AI-powered platform designed to empower students and training placement officers (TPOs) in the placement journey.
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- Uses state-of-the-art LLMs (Cohere) and Retrieval-Augmented Generation (RAG) to provide hyper-personalized assessments and roadmap generation.

Objectives:

- **To identify skill gaps and predict placement readiness** using machine learning, classifying students as Beginner, Intermediate, or Ready.
- **To generate personalized learning paths** by recommending tailored improvement strategies based on identified skill gaps.
- **To provide data-driven decision support for TPOs** through batch-level analytics and actionable insights for targeted training.
- **To automate student registration tracking and communication** by processing Excel data and sending notifications via email or WhatsApp to students and parents.
- **To streamline placement shortlisting communication** by automatically notifying selected students about placement rounds via email.

Literature Survey of the existing system

Authors	Title	Year	Publisher/Source	Methodology
Ge, M., Ricci, F., Massimo, D.	Hybrid Recommendation Systems in Education	2022	ACM RecSys Conference	Designed a hybrid recommendation model integrating content-based filtering and learner recommendations.
Patel, K., & Mehta, S.	AI-Based Personalized Learning Systems using LLMs	2023	IEEE Access	Proposed an Ai driven Personalized learning framework using LLM models to generate adaptive assesment and individual 4learning paths

Sharma, R., et al.	AI-Driven Assessment Platforms for Skill Evaluation	2023	SpringerLink	Developed an automated assessment system combining AI-based question generation and evaluation to improve skill measurement and feedback.
Zhang, L., et al.	Intelligent Student Performance Prediction using Machine Learning	2024	Elsevier Journal of Educational Data Mining	Utilized machine learning algorithms to analyze student performance data and predict academic success with high accuracy.Developed an AI-based model combining social media and ticketing data to predict event hype levels and attendance.

Limitations of existing systems

- **Static Repositories:** Most platforms use fixed databases, allowing students to predict questions over time.
- **Binary Feedback:** Existing systems only tell you if you are wrong, but Pathfinder Pal tells you why and how to fix it using AI analysis.
- **Weak Integrity Checks:** Standard platforms often ignore browser tab switches or window un focusing.
- **Disconnected Roadmaps:** Generic "Learning Paths" that don't sync with recent assessment performance, leading to redundant study of mastered topics.

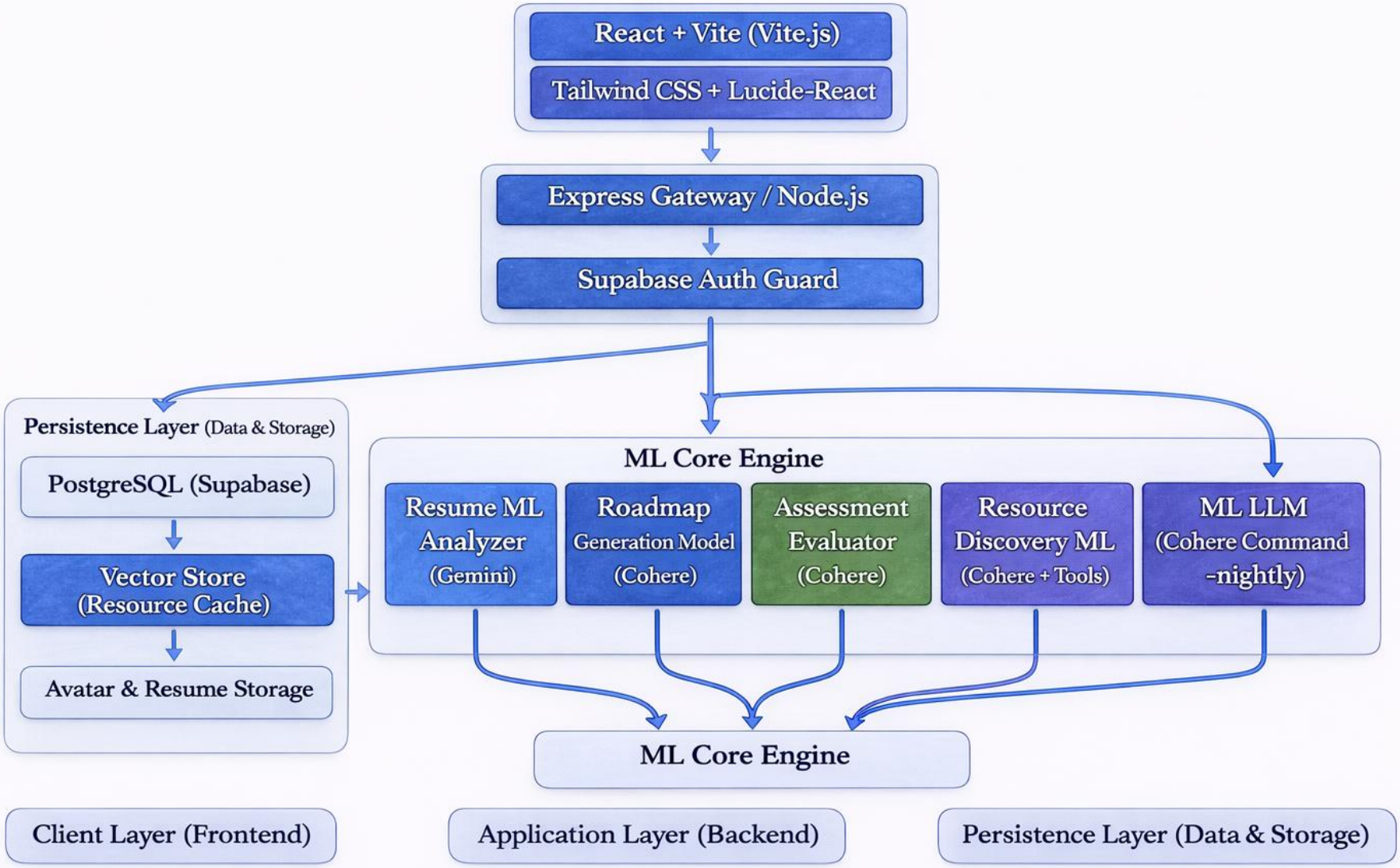
Problem statement

Students preparing for placements face generic, one-size-fits-all assessments that fail to identify their actual strengths and weaknesses.
This leads to poor interview performance, lack of targeted preparation, and low placement success rates

Solution

Pathfinder Pal analyzes performance to identify specific skill gaps and computes placement readiness, then generates a personalized learning roadmap with targeted weekly plans to help students improve efficiently and achieve better placement outcomes

System Design



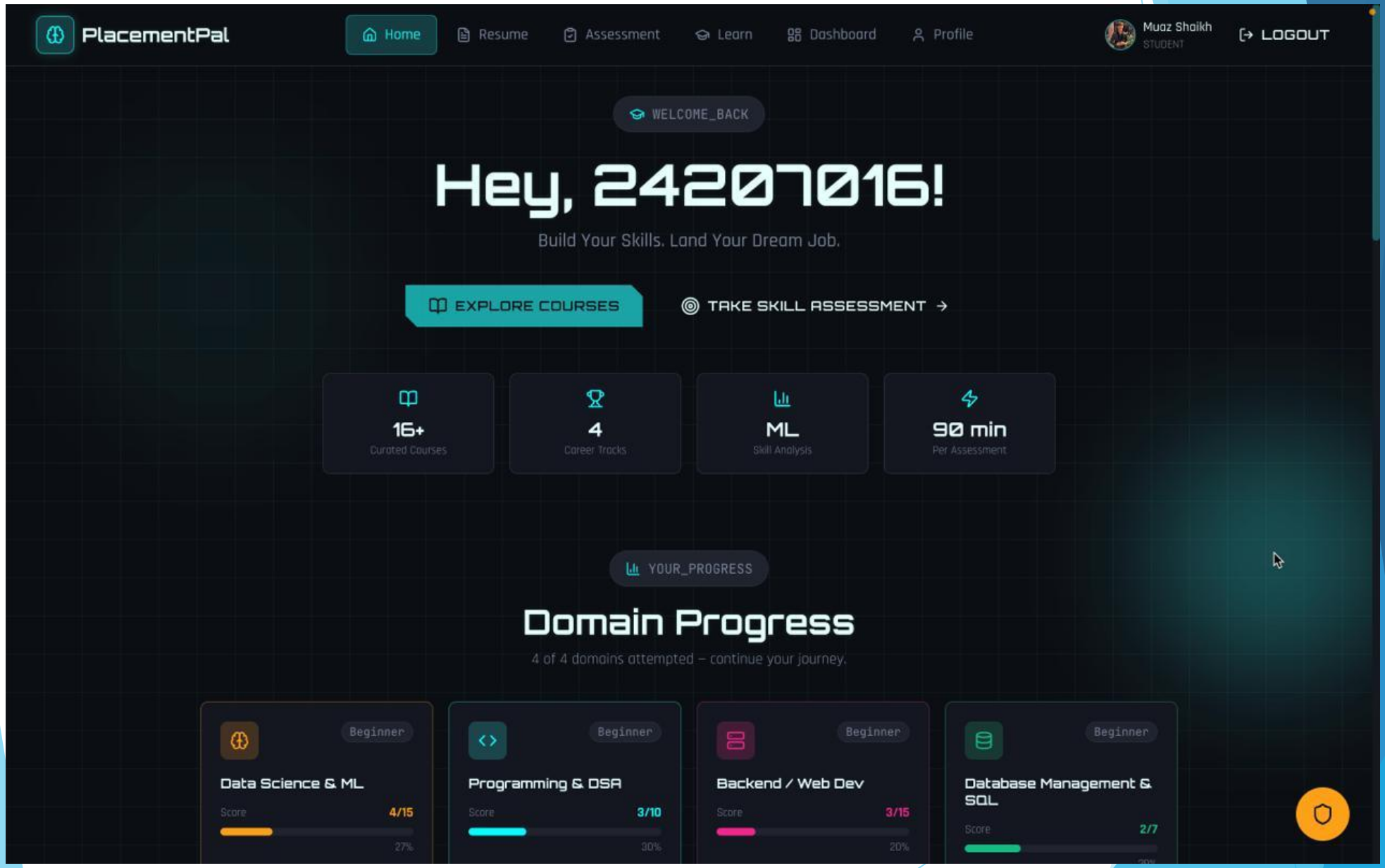
Technologies and methodologies

Frontend Framework: React (Vite), Tailwind CSS, Framer Motion, Lucide React
Builds a fast, responsive UI with smooth animations and modern design.
Backend / AP: Node.js (Express), Dotenv Handles routing, business logic, and secure environment configuration.
AI & ML Services:
Cohere Command-Nightly, RAG (Retrieval-Augmented Generation)
Enables intelligent question generation, evaluation, and domain-aware assessments.
ML Components: Evaluation Engine, Roadmap Generator, ResourceFinder
Powers assessment scoring, personalized learning plans, and smart resource discovery.
Database & Storage: PostgreSQL (Supabase), Redis (Caching)
Stores user data, assessments, and improves performance through caching.
Authentication: Supabase Auth
Provides secure user authentication and session management.
State Management: React Context API
Manages global states like user profile and authentication across the app.

Methodology:

- **Agile Development:** Implemented through iterative sprint cycles (Planning, Development, Testing).
- Focused on rapid prototyping and continuous feature integration.
- **API-First Design:** Frontend and Backend are strictly decoupled.
- **Adaptive Learning Framework:** User-centric approach where the system modifies its behavior based on performance.
- **Dynamic roadmap synthesis** using RAG (Retrieval-Augmented Generation) to fill specifically identified skill gaps.

Implementation



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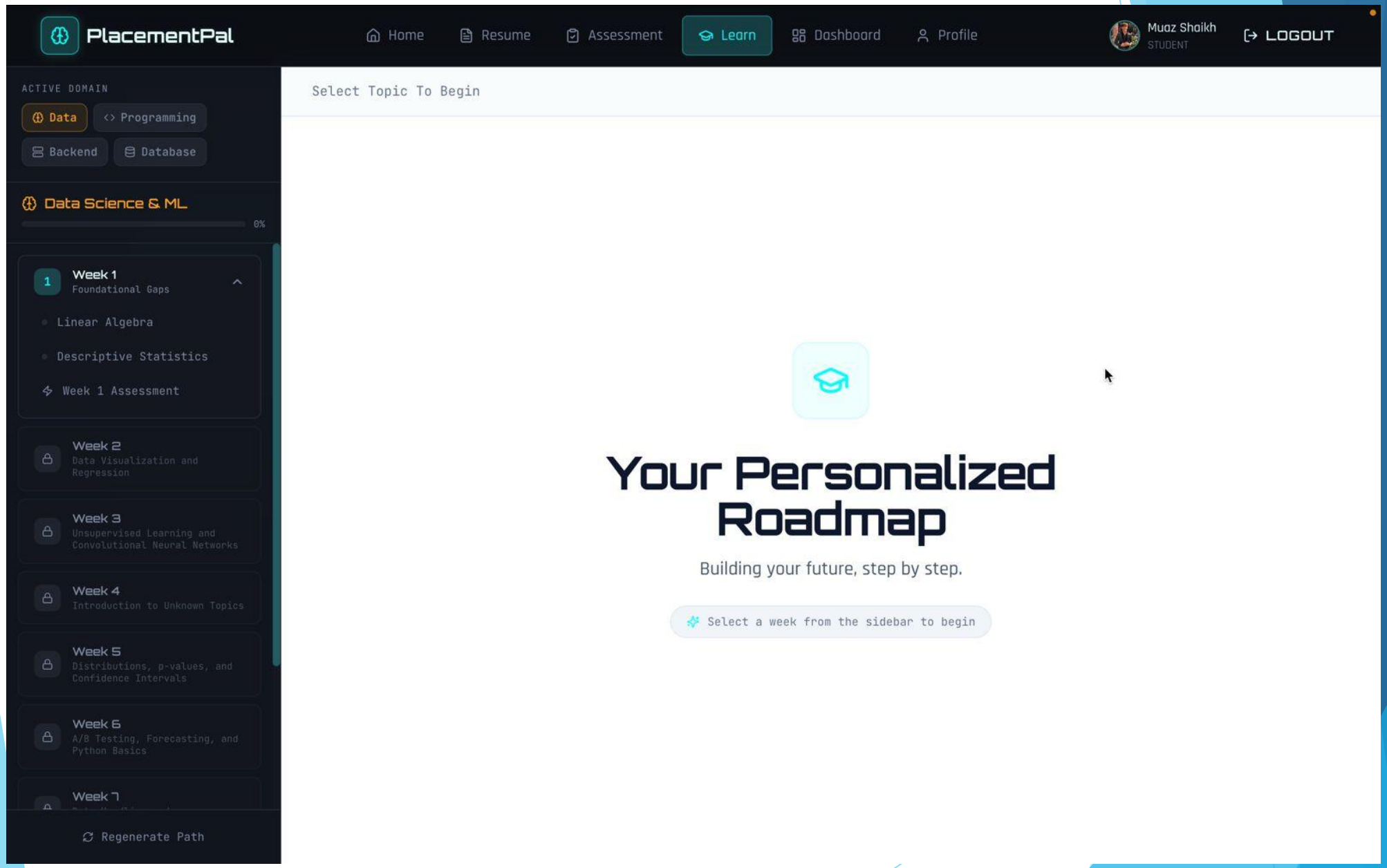

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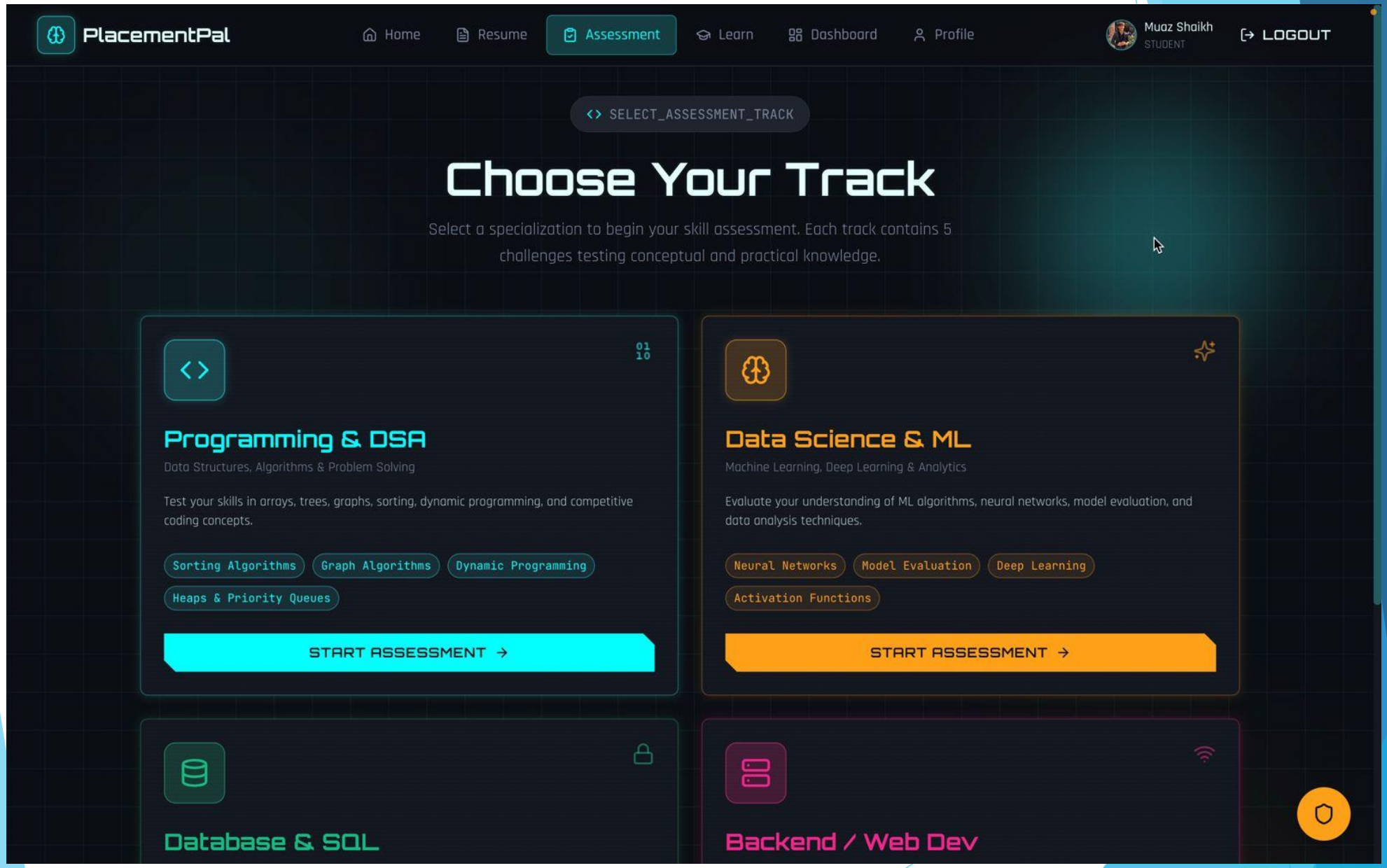

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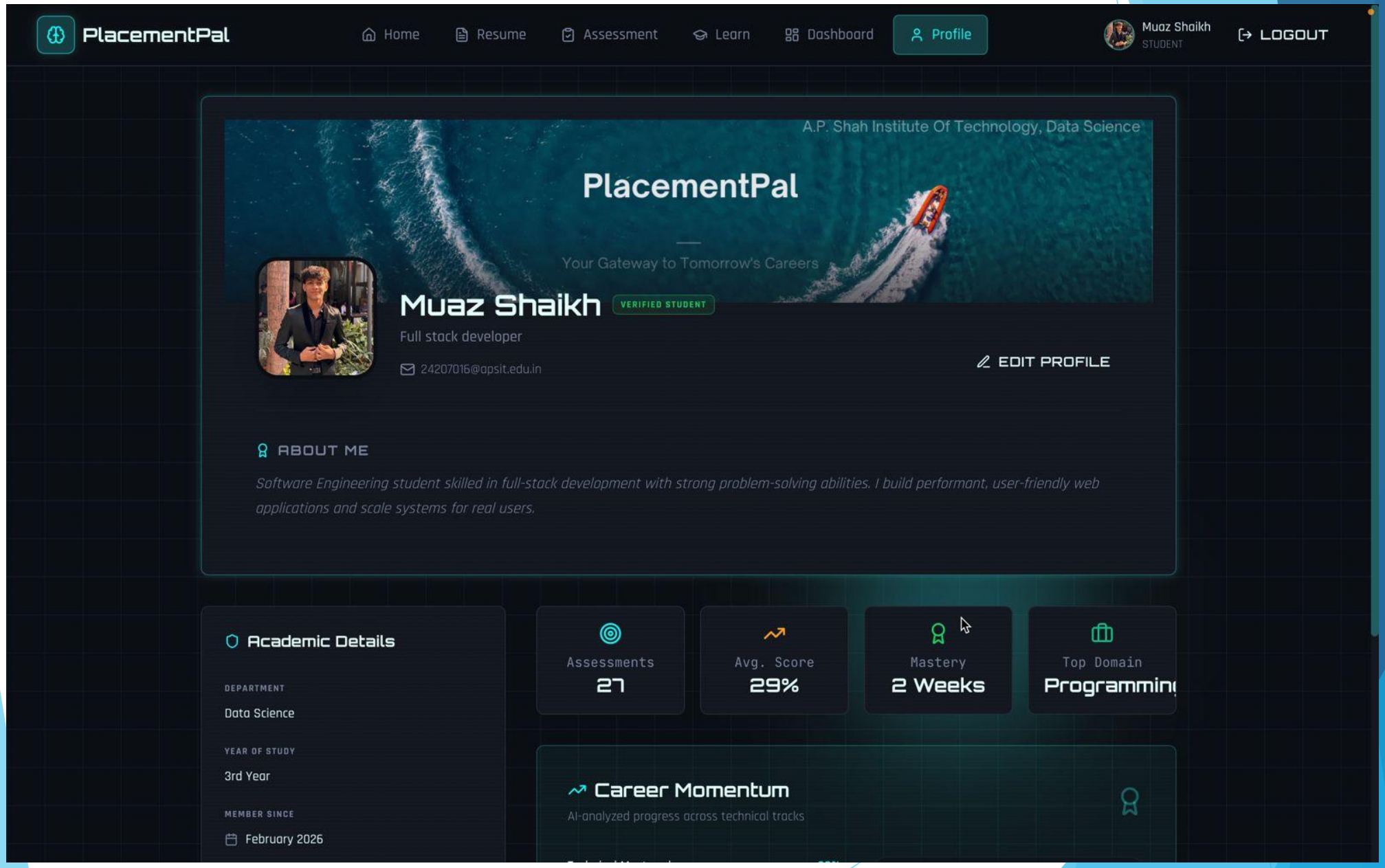

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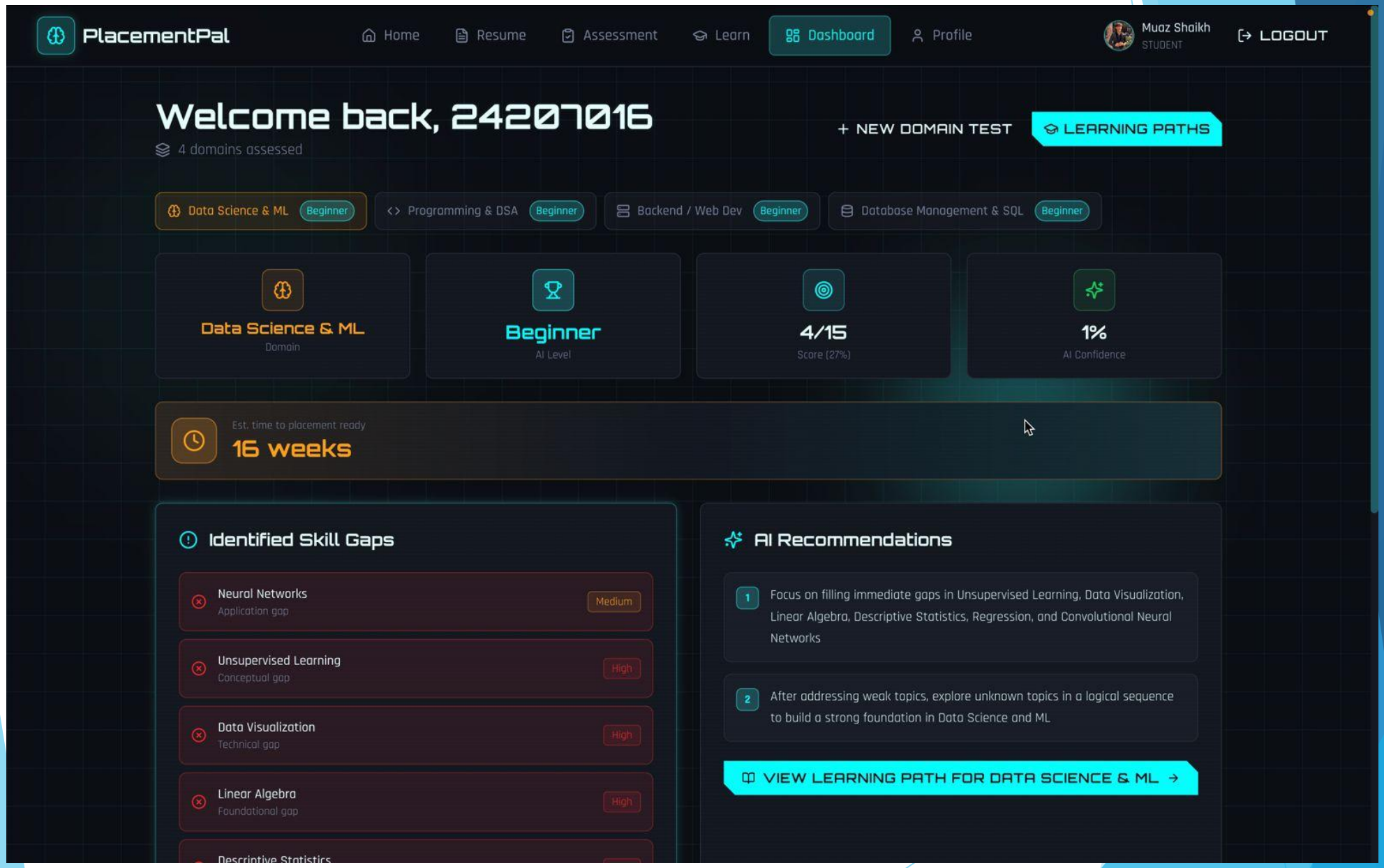

Implementation

Conclusion

- Placement Pal achieves accurate, AI-driven placement readiness assessment.
- Integrates performance data, behavioral insights, and adaptive analysis into a unified system.
- Features like readiness scores, skill gap detection, and personalized roadmaps enhance decision-making.
- A data-driven platform empowering students and TPOs with smarter placement insights.

References

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Thank You...!!